BIOMETRIC BLOOD GROUP IDENTIFICATION SYSTEM

Final Year B.E. Major Project • Deep Learning AI Analysis Report

RESEARCH / PROTOTYPE SYSTEM DISCLAIMER: This system is a research/prototype fingerprint-based blood-group classification system. It is not clinically validated and must not be used as a substitute for standard serological blood-group testing or for medical/transfusion decisions. All predictions and advisory indicators are provided solely for evaluation purposes.

1. EXAMINATION & SAMPLE AUDIT DETAILS

Report ID:	BG-000615	**Timestamp:**	2026-09-11 13:54:24
Input Source:	MFS100 Consensus (3 Impressions) | Standard Finger: LEFT THUMB	**Evaluation Status:**	LOW_CONFIDENCE
Model Name:	Champion Ensemble V2 (Consensus Soft Voting)	**Confidence:**	**35.72%**
Standard Acquisition Finger:	**LEFT THUMB** (Convention-based protocol — not automatically verified by scanner)		
Advisory Status:	NEEDS CONFIRMATION (τ=70%)	**Advisory Reason:**	**LOW_CONFIDENCE_AND_RH_UNCERTAINTY**

2. BIOMETRIC IMPRESSION & PREDICTED BLOOD GROUP



Predicted Blood Group:

O-

Confidence Score: **35.72%**
Advisory Threshold: **70%**
Advisory Screening: **NEEDS CONFIRMATION**
Derived ABO Group: **O** (71.3% prob) | Rh Factor: **-**

RESEARCH ADVISORY: Advisory threshold passed (≥ 70.0%). Note: Confidence OK does not constitute medical verification. Standard serological testing required for medical/transfusion use.

Rh FACTOR UNCERTAINTY ADVISORY: Rh factor (±) determination is uncertain (narrow Rh probability margin (0.1%); moderate 8-class confidence (35.7%); predicted Rh-negative requires serological validation due to weak ridge-Rh correlation). ABO group appears to be 'O' (71.3% confidence). Standard serological laboratory blood typing is required to confirm Rh factor.

3. 8-CLASS BLOOD GROUP PROBABILITY DISTRIBUTION

A+	A-	AB+	AB-	B+	B-	O+	O-
0.4%	22.4%	0.1%	5.9%	0.0%	0.0%	35.6%	35.7%